Majorana Modes in the Machine

Blocks 1 & 2: The Physics Bridge & Finite-Size Physics

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Outline

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2. Bulk Spectrum & Phase Transition
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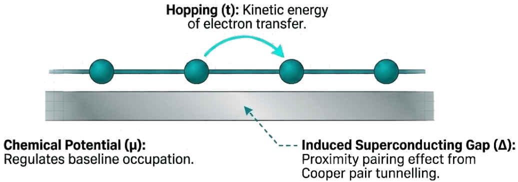
Starting point: the lattice Kitaev Hamiltonian

$$H_K = -\mu \sum_j c_j^\dagger c_j - t \sum_j (c_j^\dagger c_{j+1} + \mathrm{h.c.}) + \Delta \sum_j (c_j^\dagger c_{j+1}^\dagger + \mathrm{h.c.}) \tag{1}$$

Physical ingredients

- Spinless (or spin-polarized) fermions
- Nearest-neighbor hopping t
- p-wave pairing $\Delta \in \mathbb{R}$
- Chemical potential μ tunes filling



Questions we will answer

- When does topology emerge?
- How do edge zero modes appear?

Majorana Representation & The "Sweet Spot"

Ordinary fermions can be split into two Hermitian **Majorana operators**:

$$\gamma_{A,j} = \frac{c_j + c_j^\dagger}{\sqrt{2}}, \qquad \gamma_{B,j} = \frac{i(c_j^\dagger - c_j)}{\sqrt{2}} \tag{2}$$

By rewriting H_K in terms of Majoranas (see Appendix for details), we find a special limit:

Sweet spot: $\mu = 0$, $t = \Delta$

The intra-site coupling vanishes. Only inter-site coupling $i\gamma_{B,j}\gamma_{A,j+1}$ survives. In a chain with open boundaries, this leaves **one unpaired Majorana zero-mode at each end**.

Why is this interesting?

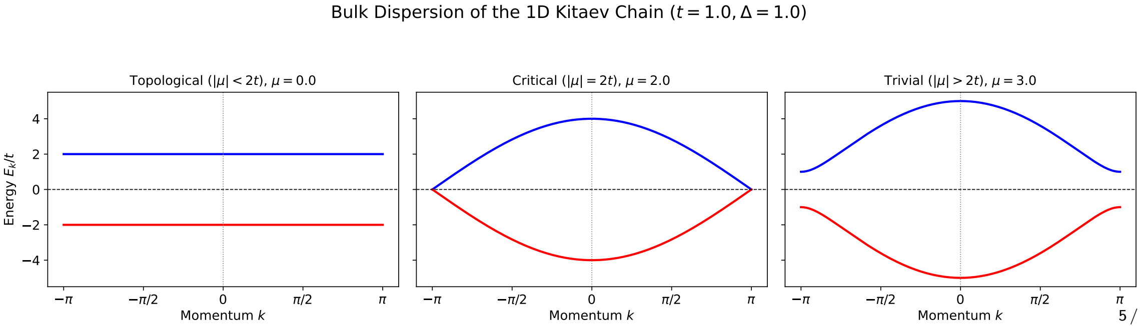
These edge Majorana modes are **robust** (protected by topology) and candidates for **fault-tolerant quantum computation**.

Bulk dispersion: topological / critical / trivial

Applying a Fourier transform and using the BdG formalism (see Appendix), the energy spectrum is given by:

$$E_k = \pm\sqrt{n_z(k)^2 + n_y(k)^2} = \pm\sqrt{(-\mu - 2t\cos k)^2 + (-2\Delta \sin k)^2} \tag{3}$$

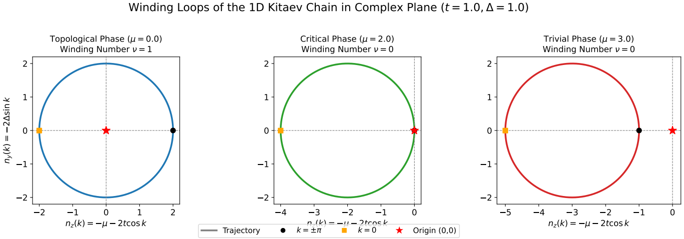
The topological phase transition occurs when the **bulk gap closes** ($E_k = 0$). This happens at $k = \{0, \pi\}$ when $\mu = \pm 2t$.



Winding number ν

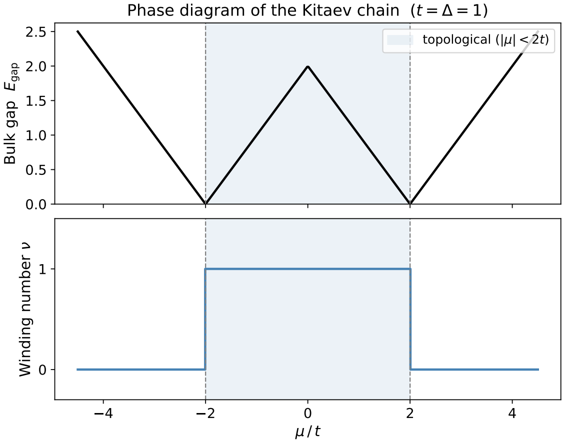
As k sweeps the BZ, the vector $\mathbf{n}(k) = (n_z, n_y)$ traces a closed loop.

$$\theta_k = \arg(n_z + i n_y), \qquad \nu = \frac{1}{2\pi}\int_{\mathrm{BZ}} \partial_k \theta_k \, dk \tag{4}$$



- $\nu = 0$: loop does *not* enclose origin $\rightarrow$ trivial
- $|\nu| = 1$: loop winds once around origin $\rightarrow$ topological

Phase diagram



Bulk gap vanishes at $\mu = \pm 2t$; winding number jumps across the transition.

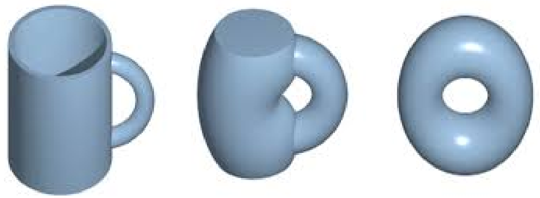
Topology: Continuity and Invariance

- **Topological Equivalence**: Two shapes are equivalent if they can be transformed into each other via "continuous deformation" (stretching, bending).
- **Invariants**: Properties that remain unchanged during deformation (e.g., number of holes / Genus).
- **Physical Link**: Tuning Hamiltonian parameters (t, μ) is essentially a "continuous deformation" of the wave function.

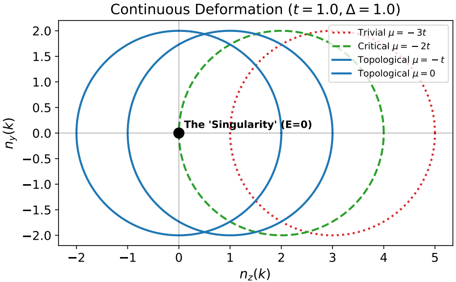
Core Intuition

You cannot turn a Sphere (0 holes) into a Donut (1 hole) without "tearing" the surface (Gap Closing).

Topological invariant: The Winding Number



with Genus 1



winding number $\nu = 1$

What is Protection?

As long as the circle doesn't cross the origin (Gap remains open), parameter fluctuations only "deform" the circle but cannot change the **integer** winding number.

Block 1 — key takeaways

1. The Kitaev Hamiltonian hosts **two phases** separated by gap closings at $\mu = \pm 2t$.
2. Majorana representation reveals the **sweet-spot** picture directly ($\mu = 0$, $t = \Delta$).
3. The BdG d-vector winding number $\nu \in \{0, 1\}$ is the **topological invariant**.

Now (Block 2): Finite size & edge modes — from bulk invariant to real-space edge states.

Step 1: Building the BdG matrix

The real-space BdG Hamiltonian is a $(2L \times 2L)$ block matrix:

$$M = \begin{pmatrix} h & d \\ -d & -h \end{pmatrix}, \quad h_{ij} = -\mu\,\delta_{ij} - t(\delta_{i,j+1} + \delta_{i+1,j}), \quad d_{ij} = \Delta(\delta_{i,j+1} - \delta_{i+1,j})$$

```python
def build_hamiltonian(self):
    h = np.zeros((L, L))
    np.fill_diagonal(h, -mu)              # onsite: chemical potential
    for j in range(L - 1):
        h[j, j+1] = h[j+1, j] = -t    # hopping

    d = np.zeros((L, L))
    for j in range(L - 1):
        d[j, j+1] =  delta                # pairing (antisymmetric)
        d[j+1, j] = -delta

    return np.block([[h, d], [-d, -h]])
```

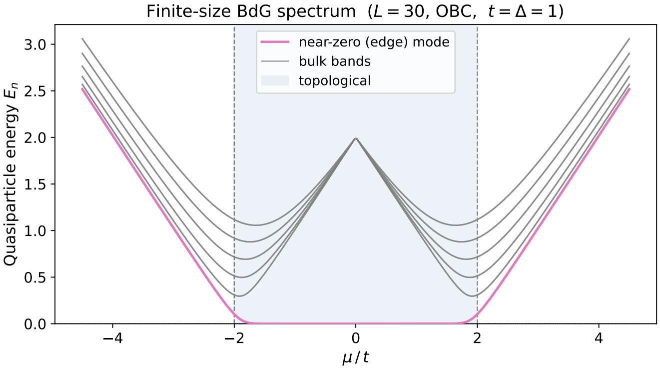
Step 2: Diagonalization & extracting the spectrum

```
def positive_spectrum(self):
    evals = np.linalg.eigvalsh(self.build_hamiltonian())
    return np.sort(np.abs(evals))[:self.L]  # see Appendix D for why
        np.abs
```

```
# Majorana splitting scan over chain lengths:
E0_arr = np.array([
    np.sort(np.abs(KitaevChain(L=l, ...).spectrum()))[0]
    for l in L_arr
])
```

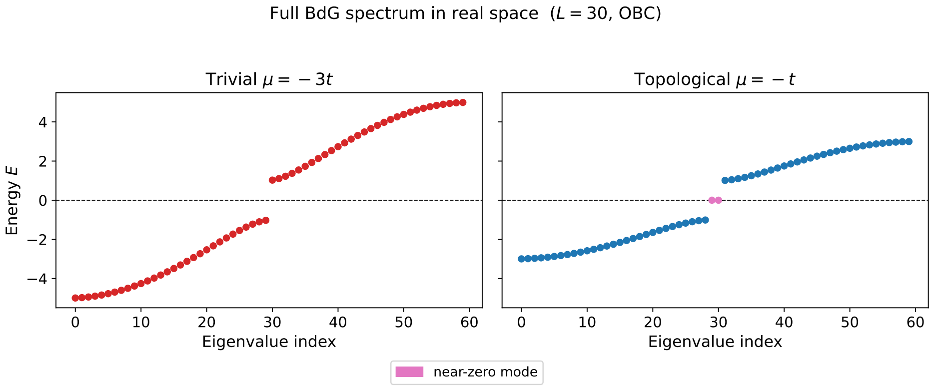
Particle-hole symmetry gives eigenvalues in $\pm E$ pairs — we take $|E|$ and keep the L smallest values to get the quasiparticle spectrum. (*Why not just filter* `evals` ≥ 0*? See Appendix D.*)

Finite-size spectrum (OBC): edge modes appear



- Pink curve: near-zero mode pinned inside the gap across the entire topological phase $|\mu| < 2t$.
- Gray curves: bulk quasiparticle bands — gapped in both phases, closing only at $\mu = \pm 2t$.

Real-space BdG snapshot



Left (trivial, $\mu = -3t$): all eigenvalues uniformly gapped — no special states.
Right (topological, $\mu = -t$): two eigenvalues pinned at $E \approx 0$ — the Majorana pair localized at opposite ends of the chain.

Exponential localization of the edge mode

At the symmetric point $t = \Delta$, the zero-mode wavefunction has an exact analytical form. Substituting the Ansatz $\psi_j \sim \lambda^j$ into the bulk BdG equations gives:

$$\lambda = \frac{\mu}{2t}, \qquad |\lambda| < 1 \text{ (topological phase)}$$

The left edge mode decays as:

$$\psi_j^L \propto \left(\frac{\mu}{2t}\right)^{j-1} = e^{-(j-1)/\xi}$$

with coherence length

$$\xi = \frac{1}{\ln(2t/|\mu|)}$$

Key points

- Pure exponential at $t = \Delta$ (no oscillations)
- $\xi \to \infty$ as $|\mu| \to 2t$ (phase boundary)
- $\xi \to 0$ as $|\mu| \to 0$ (sweet spot)

Derivation: finite-size splitting $E_0(L)$

In a **finite** chain, the left and right edge modes overlap, lifting the degeneracy. First-order perturbation theory gives:

$$\delta E = \frac{|\langle \psi^L | H | \psi^R \rangle|}{\|\psi^L\| \, \|\psi^R\|}$$

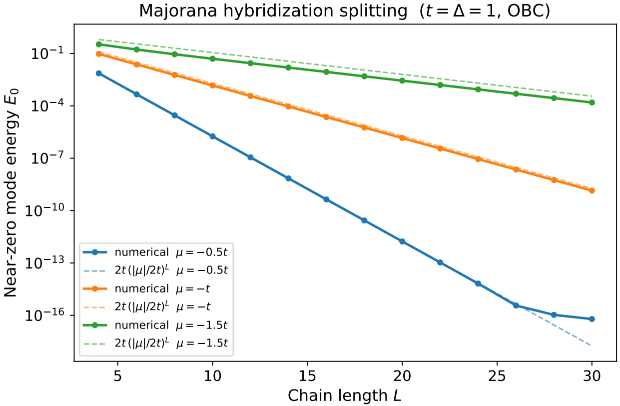
The overlap is dominated by the mid-chain region:

$$\langle \psi^L | \psi^R \rangle \sim \sum_{j=1}^{L} \lambda^{j-1} \cdot \lambda^{L-j} = L \, \lambda^{L-1}$$

Absorbing normalisation and subleading factors into a prefactor fixed by the exact sweet-spot calculation:

$$\boxed{E_0(L) \approx 2t \left(\frac{|\mu|}{2t} \right)^L}$$

Majorana hybridization splitting vs system size



Key result — at the symmetric point $t = \Delta$:

$$E_0(L) \approx 2t \left(\frac{|\mu|}{2t} \right)^L$$

- Straight lines on the semilog scale confirm **exponential** decay.
- Slope $= \ln(|\mu|/2t) = -1/\xi$ measures the **coherence length** ξ.
- Smaller $|\mu| \Rightarrow$ faster decay $\Rightarrow$ better protected zero mode.

Block 2 — key takeaways

1. The BdG matrix is built directly from μ, t, Δ — exact diagonalization gives the full spectrum with no approximations.
2. OBC spectra show **near-zero edge modes** in the topological phase — direct numerical confirmation of bulk-boundary correspondence.
3. Edge modes are **exponentially localized** with coherence length $\xi = 1/\ln(2t/|\mu|)$.
4. Majorana splitting decays as $E_0 \sim (|\mu|/2t)^L$ — exponential in L, measurable from the semilog slope.

Next session (Block 2 cont.): Jordan–Wigner transform $\rightarrow$ qubit encoding $\rightarrow$ quantum circuit for ground-state preparation.

Appendix

Appendix A: Full Majorana Representation

Recall the Majorana operators:

$$\gamma_{A,j} = \frac{c_j + c_j^\dagger}{\sqrt{2}}, \qquad \gamma_{B,j} = \frac{i(c_j^\dagger - c_j)}{\sqrt{2}} \tag{5}$$

where $\{\gamma_{\alpha,i}, \gamma_{\beta,j}\} = 2\delta_{\alpha\beta}\delta_{ij}$.

Rewriting the full Kitaev Hamiltonian in the Majorana basis yields:

$$H_K = -\mu \sum_j \left(i\gamma_{A,j}\gamma_{B,j} + \tfrac{1}{2} \right) + i \sum_j \left[(\Delta + t)\, \gamma_{B,j}\gamma_{A,j+1} + (\Delta - t)\, \gamma_{A,j}\gamma_{B,j+1} \right] \tag{6}$$

Note how setting $\mu = 0$ and $t = \Delta$ simplifies this to just the $i\gamma_{B,j}\gamma_{A,j+1}$ term.

Appendix B: Momentum Space Mapping

The tight-binding Hamiltonian in real space:

$$H = \sum_j \left[-t(c_j^\dagger c_{j+1} + \text{h.c.}) - \mu \left(c_j^\dagger c_j - \frac{1}{2} \right) + \Delta(c_j c_{j+1} + c_{j+1}^\dagger c_j^\dagger) \right] \tag{7}$$

Applying the Fourier transform $c_j = \frac{1}{\sqrt{N}} \sum_k e^{ikj} c_k$, the kinetic and chemical potential terms diagonalize trivially. The pairing term requires antisymmetrization due to fermionic statistics ($c_k c_{-k} = -c_{-k} c_k$):

$$\Delta \sum_k e^{ik} c_k c_{-k} = \Delta \sum_{k>0} (e^{ik} - e^{-ik}) c_k c_{-k} = 2i\Delta \sum_{k>0} \sin(k) c_k c_{-k} \tag{8}$$

This isolates the odd-parity p-wave superconducting nature of the model in momentum space.

Appendix C: BdG Formalism and Pseudospin

Use the Nambu spinor $\Psi_k = \begin{pmatrix} c_k \\ c_{-k}^\dagger \end{pmatrix}$ to cast H into a bilinear form:

$$H = \sum_{k>0} \Psi_k^\dagger h(k) \Psi_k, \quad h(k) = \begin{pmatrix} \xi_k & 2i\Delta \sin(k) \\ -2i\Delta \sin(k) & -\xi_k \end{pmatrix} \tag{9}$$

where the normal-state dispersion is $\xi_k = -\mu - 2t\cos(k)$.
Expanding $h(k)$ in Pauli matrices:

$$h(k) = n_y(k)\sigma^y + n_z(k)\sigma^z \implies \begin{cases} n_z(k) = -\mu - 2t\cos(k) \\ n_y(k) = -2\Delta \sin(k) \end{cases} \tag{10}$$

Diagonalizing this yields the bulk dispersion $E_k = \pm\sqrt{n_z(k)^2 + n_y(k)^2}$.

Appendix D: Why `np.abs` — the sign-filter artefact

The problem with `evals[evals ≥ 0]`

Particle-hole symmetry guarantees $\pm E$ pairs *analytically*. Numerically, near-zero Majorana modes at large L can round to tiny *negative* values — a sign filter silently drops them, and the "lowest mode" jumps to the bulk gap, creating spurious spikes.

Sign-filter artefact vs $|E|$ fix ($L = 30$, $t = \Delta = 1$, OBC)